دانشگاه اصفهان
دانشکده علوم ریاضی و آمار
نیمسال اول ۰۶-۱۴۰۵

درس: پایگاه داده
استاد: محمدرضا سعیدی

تمرین SQL

Exercise 1

Consider the following schema:

ACTOR (id, fname, lname, gender)
MOVIE (id, name, year, rank)
DIRECTORS (id, fname, lname)
CASTS (pid, mid, role)
MOVIE_DIRECTORS (did, mid)
GENRE (mid, genre)

- The id column in **ACTOR**, **MOVIE** & **DIRECTORS** tables is a key for the respective table.
- **CASTS**.pid refers to **ACTOR**.id
- **CASTS**.mid refers to **MOVIE**.id
- **MOVIE_DIRECTORS**.did refers to **DIRECTORS**.id
- **MOVIE_DIRECTORS**.mid refers to **MOVIE**.id
- **GENRE**.mid refers to **MOVIE**.id

Draw the ERD and write SQL queries for the following.

Unless otherwise specified, when asked to return all actors, directors, or movies, you should return their names (first and last, respectively). *Columns should be returned in the order listed.*

Q1.1. Draw the full attribute ERD for this system, using the Visual Paradigm tool.

Q1.2. First, read all the queries below, then use the site `https://www.filmdata.ir/` and create your database with the above schema and some data from the above site and your own information. Fill the Genre table with values like: خانوادگی، کمدی، پلیسی و ...

Q1.3. List all the cast members who are less than 40 years old whose names begin with any of the five letters of the Persian alphabet.

Q1.4. List all the directors who directed a movie of genre 'خانوادگی' in a leap year. (You need to check the genre and that the year is divisible by 4.) Return the director's name along with the movie name and movie year.

Q1.5. List the names of all the actors who played in the movie 'زیبا صدایم کن' made in 1403.

Q1.6. List all the actors who acted in a film before 1390 or in a film after 1400.

Q1.7. List all directors who directed 3 movies or more, in descending order of the number of movies they directed. List the director names and the number of movies each directed.

Q1.8. We want to find actors that played two or more roles in the same movie. Notice that Casts may have occasional duplicates, but we are not interested in these: we want actors that had two or more distinct roles in the same movie.

a. Write a query that returns the actors' names, the movie name, and the number of distinct roles that they played in that movie (which will be at least 2).

b. Write a query that returns the actors' names, the movie name, and all the distinct roles (two or more) that the actor played in that movie.

Q1.9. (a) For each year, count the number of movies in that year that had only female actors. (b) Now make a small change: for each year, report the percentage of movies with only female actors made that year, and also the total number of movies made that year. For example, one answer will be:

```
1399  31.80  200
```

meaning that in 1399 there were 200 movies, and 31.80% had only female actors.

Q1.10. Find the film(s) with the largest cast. Return the movie title and the size of the cast. By "cast size" we mean the number of distinct actors that played in that movie: if an actor played multiple roles, or if it simply occurs multiple times in casts, we still count her/him only once.

Q1.11. A decade is a sequence of 10 consecutive years. For example, 1395..1404 is a decade. Find and return the decade with the largest number of films.

Hint: this query combines several subtleties. One thing you may want to check is that the number of movies in that decade doesn't exceed the total number of movies in the database.

Q1.12. Find the actors who have played in a film of genre 'خانوادگی' and whose age is higher than the average age of all male actors.

Q1.13. The Bacon number of an actor is the length of the shortest path between the actor and Kevin Bacon in the "co-acting" graph. That is, Kevin Bacon has Bacon number 0; all actors who acted in the same film as KB have Bacon number 1; all actors who acted in the same film as some actor with Bacon number 1 have Bacon number 2, etc. Return all actors whose Bacon number is 2. Use 'علی نصیریان' instead of Bacon.

Q1.14. List all actors who play in just one genre of movies.

Q1.15. All actors who play in two roles with one director.

Q1.16. Actors who play more genres than the directors that they worked with them.

To ponder: how can you compute the Bacon number of each actor in the database?